



The desktop processor segment saw quite the upheaval last year with the launch of Intel's 12th Gen Alder Lake processors featuring their new heterogeneous microarchitecture. AMD continues to experiment with newer packaging technologies such as wafer-stacking resulting in the Ryzen 7 5800X3D that beats a lot of the mainstream processors in gaming workloads thanks to a massive cache that's placed really close to the cores.

Intel came out with their 13th Gen Raptor Lake processors featuring several tweaks and improvements that resulted in a surprisingly high gain in performance over their previous gen processors and AMD also overhauled a significant chunk of their silicon design to get even more performance. This year, workloads continued to be dominated by single-threaded performance and AMD seems to have worked a lot in this space to inch closer to Intel's single-threaded performance lead.

#### Winner: AMD Ryzen 9 7950X

AMD always had the lead in core counts but the single-threaded performance of said cores were always lagging behind Intel. So AMD always had the lead in multi-threaded performance

but in gaming and other workloads that are driven by single-threaded performance, Intel was always ahead. The architectural improvements with Zen 4 have resulted in the Ryzen processors gaining a lot in terms of single-threaded performance and coupled with the massive core count of the flagship Ryzen 7000 processor, the AMD Ryzen 9 7950X managed to bag the Zero1 Award this year. We suspect that if the Intel 13900K had a few more performance cores instead of efficiency cores, then the tables would have easily turned and Intel would have been this year's winner.

#### Runner-up: Intel Core i9-13900K

Intel really outdid itself with the Raptor Lake launch. Doubling the E-Cores

appears to have given the 13th Gen processors a significant boost in performance which is quite unexpected. The performance gains were indicative of what are usually seen with completely new architectures. However, with the launch of the Ryzen 7000 processors, Intel lost a decent chunk of the single-threaded performance lead that it had enjoyed for most of this decade. Sure, throwing in a lot of cores allows you to gain more multi-threaded performance but you can only add so much to a desktop processor.

#### Best Buy: Intel Core i5-13600K

While the AMD Ryzen 5 7600X is a great processor on several fronts - power consumption, multi-threaded performance and single-threaded performance, the Intel Core i5-13600K is no different. The decision came down to which of the two became the better buy. Simply looking at the processors, the AMD Ryzen 5 7600X retails for about ₹5K cheaper than the 13600K, but the 13600K had a lead of about 5-8 per cent in single-threaded performance. However, there was another key change this year. AMD shifted to the AM5 socket with the advent of the Ryzen 7000 family and that resulted in a significant bump in cost of ownership. The cheapest board capable of driving the 13600K comes at around ₹7,200 whereas the cheapest AM5 motherboard starts off at ₹17,750. The price difference between the processors is easily outweighed by the price difference of getting a new motherboard, and that tilted the scales in Intel's favour.

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#### DIGIT BEST BUY WINNER-2022



**Intel Core i9-13900K**  
Price: ₹58,350



**Intel Core i5-13600K**  
Price: ₹31,600

# Processors

## Big cores vs small cores



**AMD Ryzen 9 7950X**  
Price: ₹57,955